EDA

```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(lubridate)
library(forecast)
library(knitr)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

Load the Harmonized Data and a First Glimpse

```r
macro_df <- read_csv("../data/macro_panel_wide_raw.csv", show_col_types = FALSE) |>
  mutate(quarterly_date = as.Date(quarterly_date)) |>
  mutate(household_consumption_pc = household_consumption / population * 1000000,
         gross_domestic_income_pc = gross_domestic_income / population * 1000000,
         gross_private_domestic_investment_pc = gross_private_domestic_investment / population * 1000000
  arrange(quarterly_date)

glimpse(macro_df)
```

```
## Rows: 197
## Columns: 14
## $ quarterly_date                      <date> 1977-01-01, 1977-04-01, 1977-07-~
## $ population                          <dbl> 23591713, 23656739, 23725843, 238~
## $ national_net_savings                <dbl> 15180, 14684, 13724, 14136, 13512~
## $ household_net_savings               <dbl> 13632, 17048, 13920, 15024, 19168~
## $ corporate_net_savings               <dbl> 4660, 3260, 3592, 4984, 3892, 350~
## $ gross_domestic_income               <dbl> 219912, 224808, 229724, 235960, 2~
## $ household_consumption               <dbl> 418612, 417567, 424081, 428064, 4~
## $ unemployment_rate                   <dbl> 7.8, 7.8, 8.3, 8.5, 8.5, 8.4, 8.4~
## $ cpi_inflation_indicator             <dbl> 32.8, 33.5, 34.2, 34.9, 35.7, 36.~
## $ market_interest_rates               <dbl> 8.164062, 7.700000, 7.500000, 7.5~
## $ gross_private_domestic_investment   <dbl> 135625, 135843, 133541, 132462, 1~
```

```
## $ household_consumption_pc             <dbl> 17744.03, 17651.08, 17874.22, 179~
## $ gross_domestic_income_pc             <dbl> 9321.578, 9502.916, 9682.438, 991~
## $ gross_private_domestic_investment_pc <dbl> 5748.841, 5742.254, 5628.504, 556~
```

```r
summary(macro_df)
```

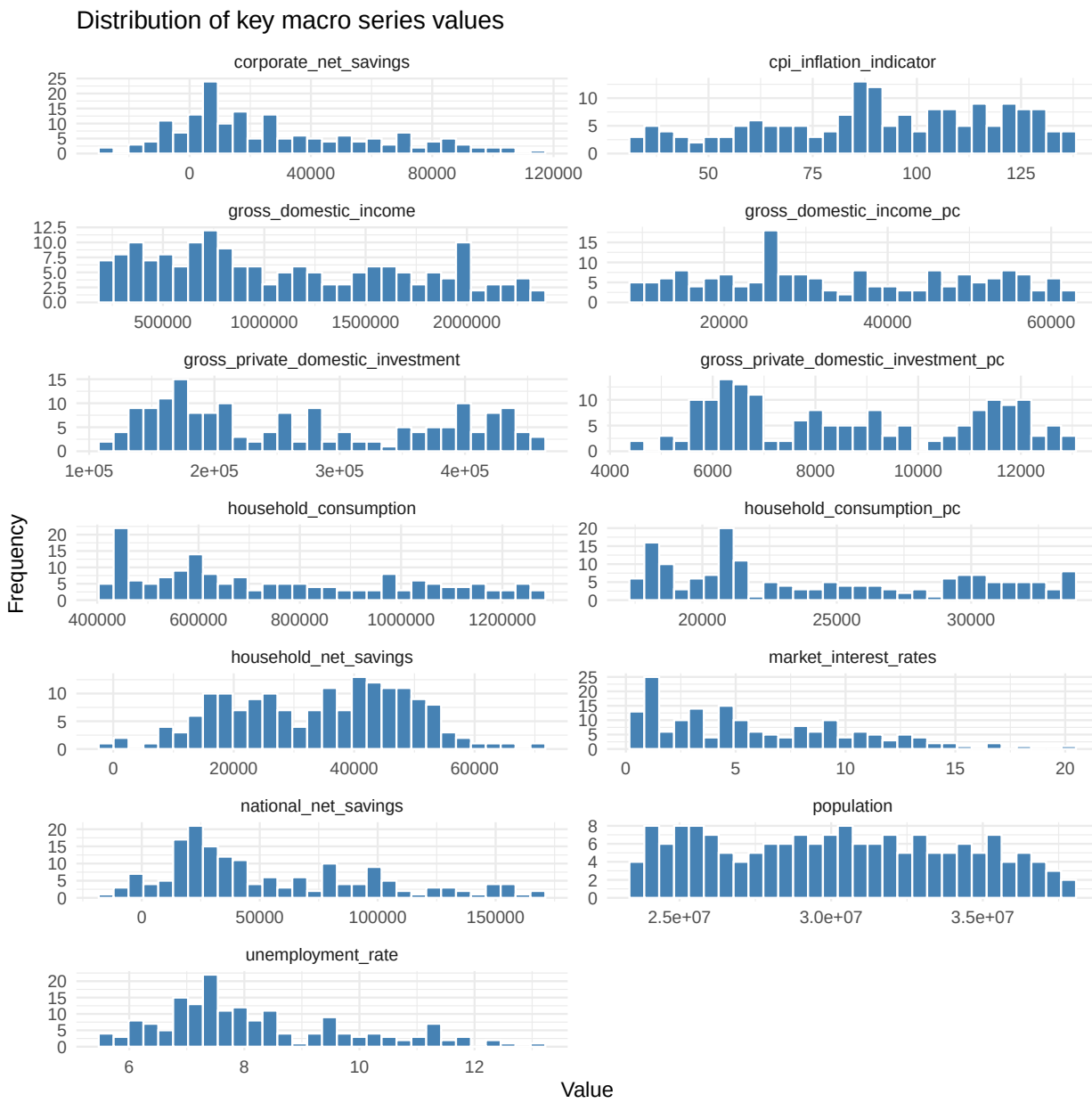
```
##  quarterly_date         population       national_net_savings
##  Min.   :1977-01-01   Min.   :23591713   Min.   :-20116
##  1st Qu.:1989-04-01   1st Qu.:27118921   1st Qu.: 23627
##  Median :2001-07-01   Median :30965926   Median : 44448
##  Mean   :2001-07-01   Mean   :31320721   Mean   : 66616
##  3rd Qu.:2013-10-01   3rd Qu.:35113143   3rd Qu.: 99984
##  Max.   :2026-01-01   Max.   :41651653   Max.   :274012
##                       NA's   :1          NA's   :1
##  household_net_savings corporate_net_savings gross_domestic_income
##  Min.   : -1168        Min.   :-27516        Min.   : 219912
##  1st Qu.: 23350        1st Qu.:  6463        1st Qu.: 666384
##  Median : 37636        Median : 25036        Median :1144364
##  Mean   : 41686        Mean   : 35195        Mean   :1314245
##  3rd Qu.: 48448        3rd Qu.: 61384        3rd Qu.:1917602
##  Max.   :384240        Max.   :187928        Max.   :3278568
##  NA's   :1             NA's   :1             NA's   :1
##  household_consumption unemployment_rate cpi_inflation_indicator
##  Min.   : 417567       Min.   : 4.900    Min.   : 32.80
##  1st Qu.: 570258       1st Qu.: 6.900    1st Qu.: 74.90
##  Median : 761540       Median : 7.600    Median : 98.60
##  Mean   : 826977       Mean   : 8.038    Mean   : 98.83
##  3rd Qu.:1087645       3rd Qu.: 9.100    3rd Qu.:123.30
##  Max.   :1431258       Max.   :13.100    Max.   :165.00
##  NA's   :1
##  market_interest_rates gross_private_domestic_investment
##  Min.   : 0.500        Min.   :113609
##  1st Qu.: 1.936        1st Qu.:178025
##  Median : 4.674        Median :277918
##  Mean   : 5.532        Mean   :290796
##  3rd Qu.: 8.569        3rd Qu.:405561
##  Max.   :20.153        Max.   :489561
##                        NA's   :1
##  household_consumption_pc gross_domestic_income_pc
##  Min.   :17564            Min.   : 9322
##  1st Qu.:20622            1st Qu.:24572
##  Median :24623            Median :36987
##  Mean   :25540            Mean   :39013
##  3rd Qu.:31086            3rd Qu.:54106
##  Max.   :34528            Max.   :78858
##  NA's   :1                NA's   :1
##  gross_private_domestic_investment_pc
##  Min.   : 4523
##  1st Qu.: 6637
##  Median : 8926
##  Mean   : 8965
##  3rd Qu.:11272
##  Max.   :12928
##  NA's   :1
```

```r
na_summary <- macro_df |>
  summarise(across(-quarterly_date, ~ sum(is.na(.)))) |>
  pivot_longer(
    cols = everything(),
    names_to = "series",
    values_to = "n_missing"
  ) |>
  arrange(desc(n_missing))

na_summary
```

```
## # A tibble: 13 x 2
##    series                                n_missing
##    <chr>                                     <int>
##  1 population                                    1
##  2 national_net_savings                          1
##  3 household_net_savings                         1
##  4 corporate_net_savings                         1
##  5 gross_domestic_income                         1
##  6 household_consumption                         1
##  7 gross_private_domestic_investment             1
##  8 household_consumption_pc                      1
##  9 gross_domestic_income_pc                      1
## 10 gross_private_domestic_investment_pc          1
## 11 unemployment_rate                             0
## 12 cpi_inflation_indicator                       0
## 13 market_interest_rates                         0
```

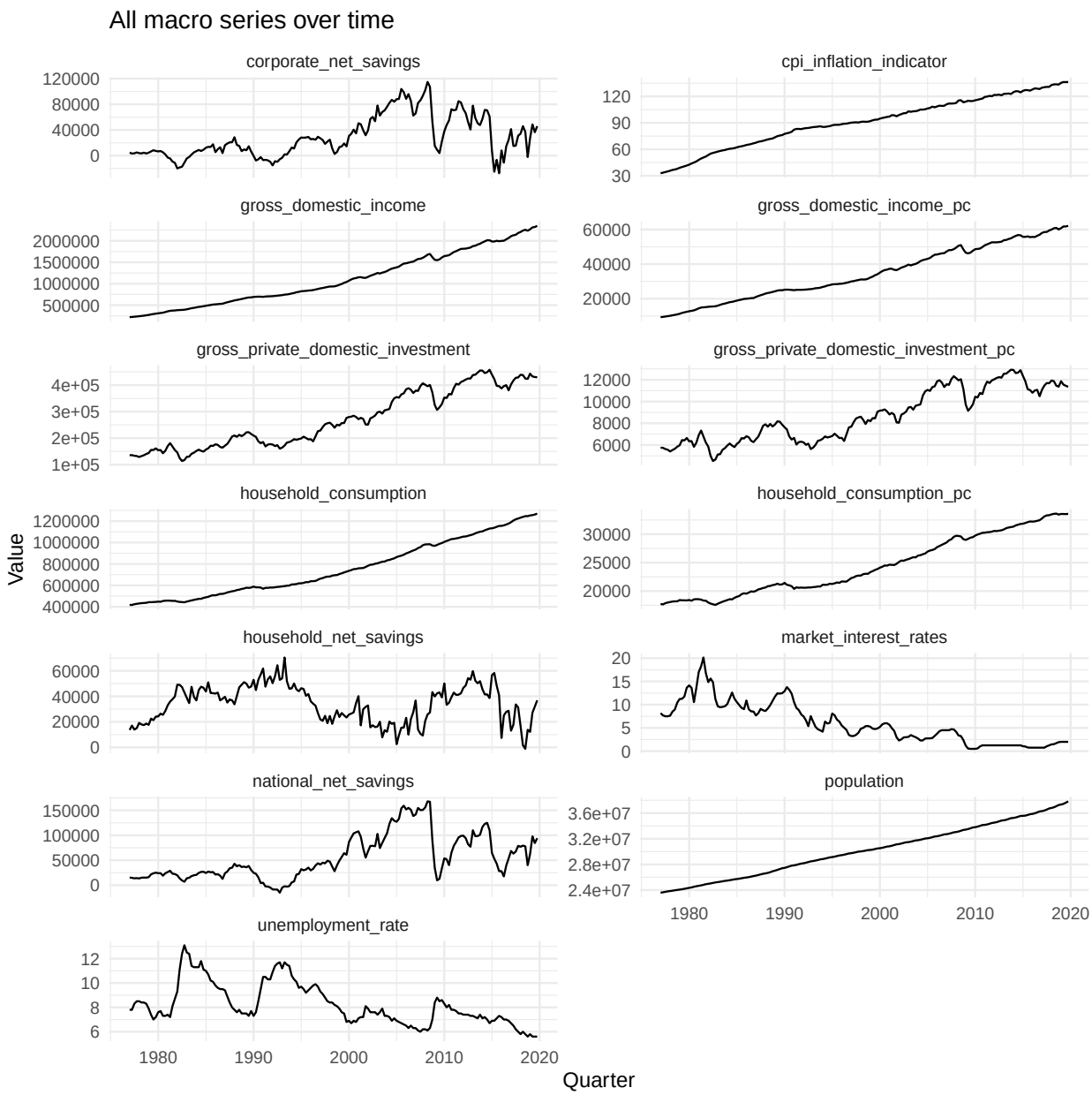
```r
# Remove the NA values at the end
macro_df <- macro_df |>
  filter(quarterly_date <= as.Date("2019-12-31"))
```

```r
key_vars <- macro_df |>
  select(-quarterly_date) |>
  pivot_longer(
    cols = everything(),
    names_to = "series",
    values_to = "value"
  )

# Adjust plot size for better readability
ggplot(key_vars, aes(x = value)) +
  geom_histogram(bins = 30, fill = "steelblue", color = "white") +
  facet_wrap(~ series, scales = "free", ncol = 2)+
  theme_minimal() +
  ggtitle("Distribution of key macro series values") +
  xlab("Value") +
  ylab("Frequency")
```




```r
macro_long <- macro_df |>
  pivot_longer(
    cols = -quarterly_date,
    names_to = "series",
    values_to = "value"
  )

ggplot(macro_long, aes(x = quarterly_date, y = value)) +
  geom_line() +
  facet_wrap(~ series, scales = "free_y", ncol = 2) +
  theme_minimal() +
  labs(
    title = "All macro series over time",
    x = "Quarter",
```

```
    y = "Value"
  )
```




```
eda_df <- macro_df |>
  select(
    quarterly_date,
    population,
    national_net_savings,
    household_net_savings,
    corporate_net_savings,
    gross_domestic_income,
    household_consumption,
    unemployment_rate,
    cpi_inflation_indicator,
```

```r
    market_interest_rates,
    gross_private_domestic_investment
  ) |>
  rename(
    date = quarterly_date,
    pop = population,
    nat_save = national_net_savings,
    hh_save = household_net_savings,
    corp_save = corporate_net_savings,
    gdi = gross_domestic_income,
    cons = household_consumption,
    unemp = unemployment_rate,
    cpi = cpi_inflation_indicator,
    interest = market_interest_rates,
    gpdi = gross_private_domestic_investment
  ) |>
  mutate( # Standardize unit
    hh_rate = (hh_save / gdi) ,
    nat_rate = (nat_save / gdi)
  ) |>
  arrange(date)

glimpse(eda_df)
```

```
## Rows: 172
## Columns: 13
## $ date      <date> 1977-01-01, 1977-04-01, 1977-07-01, 1977-10-01, 1978-01-01,~
## $ pop       <dbl> 23591713, 23656739, 23725843, 23801445, 23849503, 23903687, ~
## $ nat_save  <dbl> 15180, 14684, 13724, 14136, 13512, 14992, 15232, 15124, 1603~
## $ hh_save   <dbl> 13632, 17048, 13920, 15024, 19168, 17984, 17416, 18660, 1766~
## $ corp_save <dbl> 4660, 3260, 3592, 4984, 3892, 3504, 4576, 3336, 4588, 6492, ~
## $ gdi       <dbl> 219912, 224808, 229724, 235960, 241320, 248684, 255204, 2630~
## $ cons      <dbl> 418612, 417567, 424081, 428064, 431165, 434052, 436086, 4373~
## $ unemp     <dbl> 7.8, 7.8, 8.3, 8.5, 8.5, 8.4, 8.4, 8.3, 7.9, 7.4, 7.0, 7.2, ~
## $ cpi       <dbl> 32.8, 33.5, 34.2, 34.9, 35.7, 36.6, 37.1, 37.9, 39.0, 39.9, ~
## $ interest  <dbl> 8.164062, 7.700000, 7.500000, 7.500000, 7.630769, 8.492308, ~
## $ gpdi      <dbl> 135625, 135843, 133541, 132462, 128906, 132350, 135470, 1404~
## $ hh_rate   <dbl> 0.06198843, 0.07583360, 0.06059445, 0.06367181, 0.07942980, ~
## $ nat_rate  <dbl> 0.06902761, 0.06531796, 0.05974125, 0.05990846, 0.05599204, ~
```

```r
summary(eda_df)
```

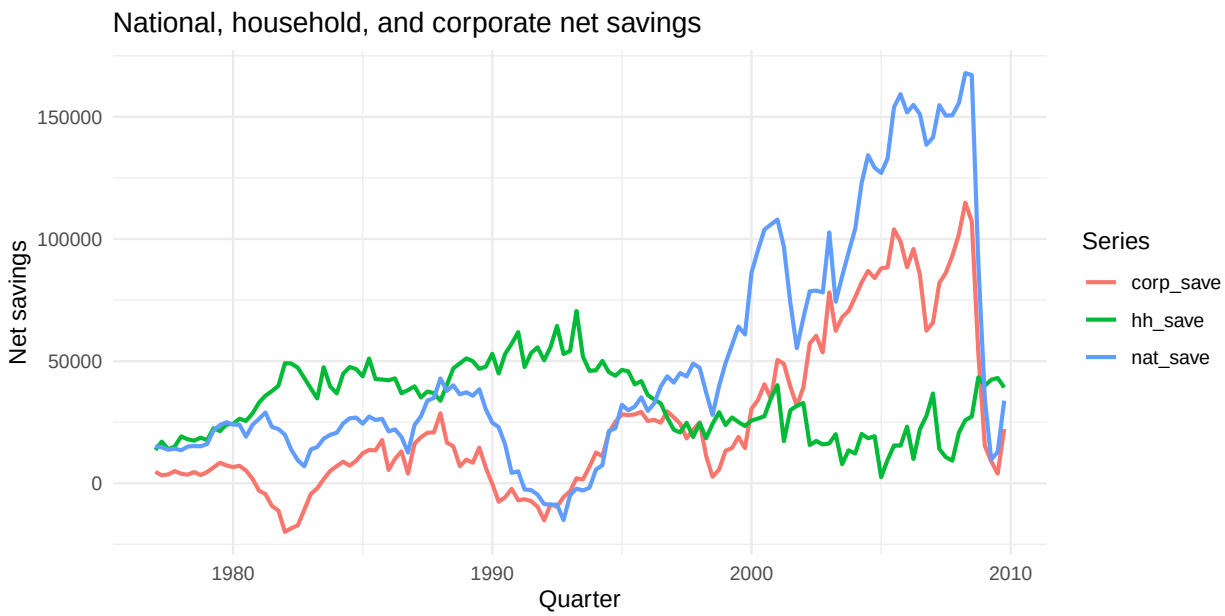
```
##       date                 pop              nat_save         hh_save
##  Min.   :1977-01-01   Min.   :23591713   Min.   :-15092   Min.   :-1168
##  1st Qu.:1987-09-08   1st Qu.:26518428   1st Qu.: 21934   1st Qu.:22080
##  Median :1998-05-16   Median :30117676   Median : 39424   Median :35974
##  Mean   :1998-05-17   Mean   :30161946   Mean   : 54932   Mean   :34030
##  3rd Qu.:2009-01-23   3rd Qu.:33448892   3rd Qu.: 85037   3rd Qu.:45850
##  Max.   :2019-10-01   Max.   :37828162   Max.   :167916   Max.   :70500
##    corp_save           gdi                cons              unemp
##  Min.   :-27516   Min.   : 219912   Min.   : 417567   Min.   : 5.600
##  1st Qu.:  4903   1st Qu.: 592240   1st Qu.: 543798   1st Qu.: 7.100
##  Median : 18592   Median : 937042   Median : 692971   Median : 7.700
##  Mean   : 28500   Mean   :1103727   Mean   : 757738   Mean   : 8.219
```

```
##  3rd Qu.: 50499   3rd Qu.:1623505   3rd Qu.: 983645   3rd Qu.: 9.400
##  Max.   :114844   Max.   :2350184   Max.   :1270170   Max.   :13.100
##       cpi            interest           gpdi           hh_rate
##  Min.   : 32.80   Min.   : 0.500   Min.   :113609   Min.   :-0.0005171
##  1st Qu.: 69.45   1st Qu.: 2.000   1st Qu.:173278   1st Qu.: 0.0195790
##  Median : 91.25   Median : 4.857   Median :250848   Median : 0.0295227
##  Mean   : 90.99   Mean   : 5.944   Mean   :270653   Mean   : 0.0456201
##  3rd Qu.:114.88   3rd Qu.: 9.048   3rd Qu.:380890   3rd Qu.: 0.0739474
##  Max.   :136.40   Max.   :20.153   Max.   :458166   Max.   : 0.1282702
##     nat_rate
##  Min.   :-0.02075
##  1st Qu.: 0.03400
##  Median : 0.05131
##  Mean   : 0.04962
##  3rd Qu.: 0.06357
##  Max.   : 0.10841
```

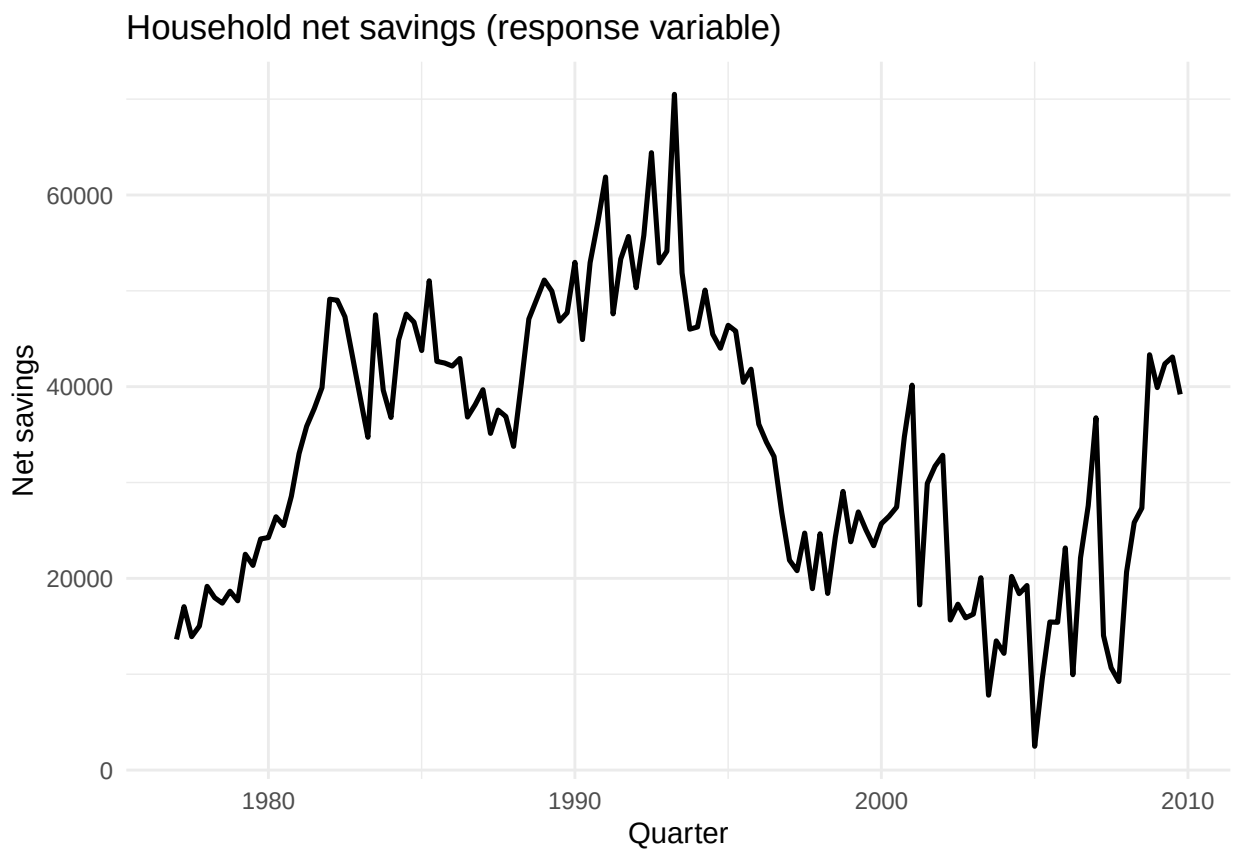
```r
saveRDS(eda_df, file = "../output/eda_df_nodiff.rds")
```

```r
# We keep the last 5 years (20 quarters) for testing, which means the training set includes data up to
train_df <- eda_df |>
  filter(date <= as.Date("2009-12-31"))
test_df <- eda_df |>
  filter(date > as.Date("2009-12-31"))

saveRDS(train_df, file = "../output/train_df.rds")
saveRDS(test_df, file = "../output/test_df.rds")
```

```r
# Reload train data to ensure we are using the same dataset as in the later modeling steps
eda_df <- readRDS("../output/train_df.rds")

savings_long <- eda_df |>
  select(date, nat_save, hh_save, corp_save) |>
  pivot_longer(
    cols = -date,
    names_to = "series",
    values_to = "value"
  )

ggplot(savings_long, aes(x = date, y = value, color = series)) +
  geom_line(linewidth = 0.9) +
  theme_minimal() +
  labs(
    title = "National, household, and corporate net savings",
    x = "Quarter",
    y = "Net savings",
    color = "Series"
  )
```

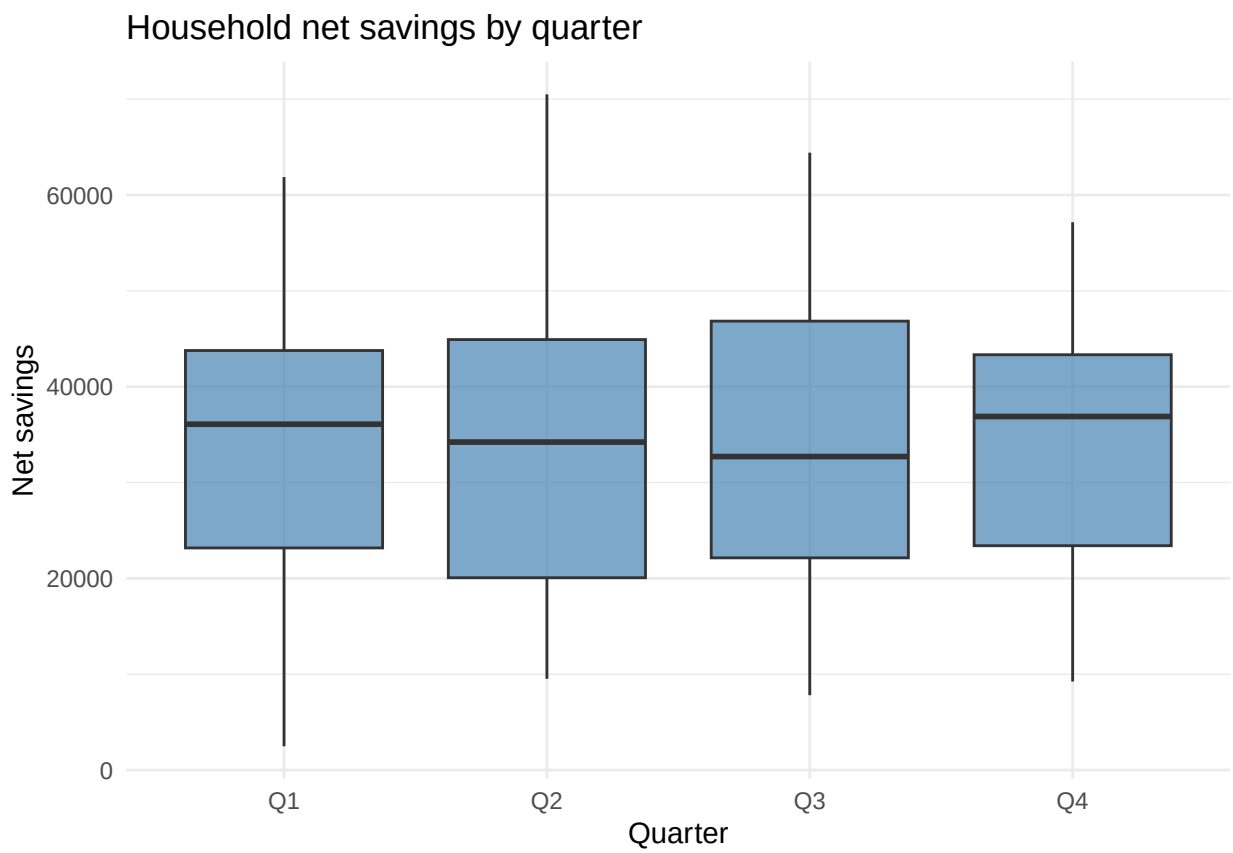



```r
ggplot(eda_df, aes(x = date, y = hh_save)) +
  geom_line(linewidth = 0.9) +
  theme_minimal() +
  labs(
    title = "Household net savings (response variable)",
    x = "Quarter",
    y = "Net savings"
  )
```



Addition_1:

```
eda_df <- eda_df |>
  mutate(
    quarter_lab = factor(quarter(date), labels = c("Q1", "Q2", "Q3", "Q4"))
  )

ggplot(eda_df, aes(x = quarter_lab, y = hh_save)) +
  geom_boxplot(fill = "steelblue", alpha = 0.7) +
  theme_minimal() +
  labs(
    title = "Household net savings by quarter",
    x = "Quarter",
    y = "Net savings"
  )
```




```
quarter_summary <- eda_df |>
  group_by(quarter_lab) |>
  summarise(
    mean_hh_save = mean(hh_save, na.rm = TRUE),
    sd_hh_save = sd(hh_save, na.rm = TRUE),
    .groups = "drop"
  )

kable(quarter_summary, digits = 1,
      caption = "Quarterly summary of household net savings")
```

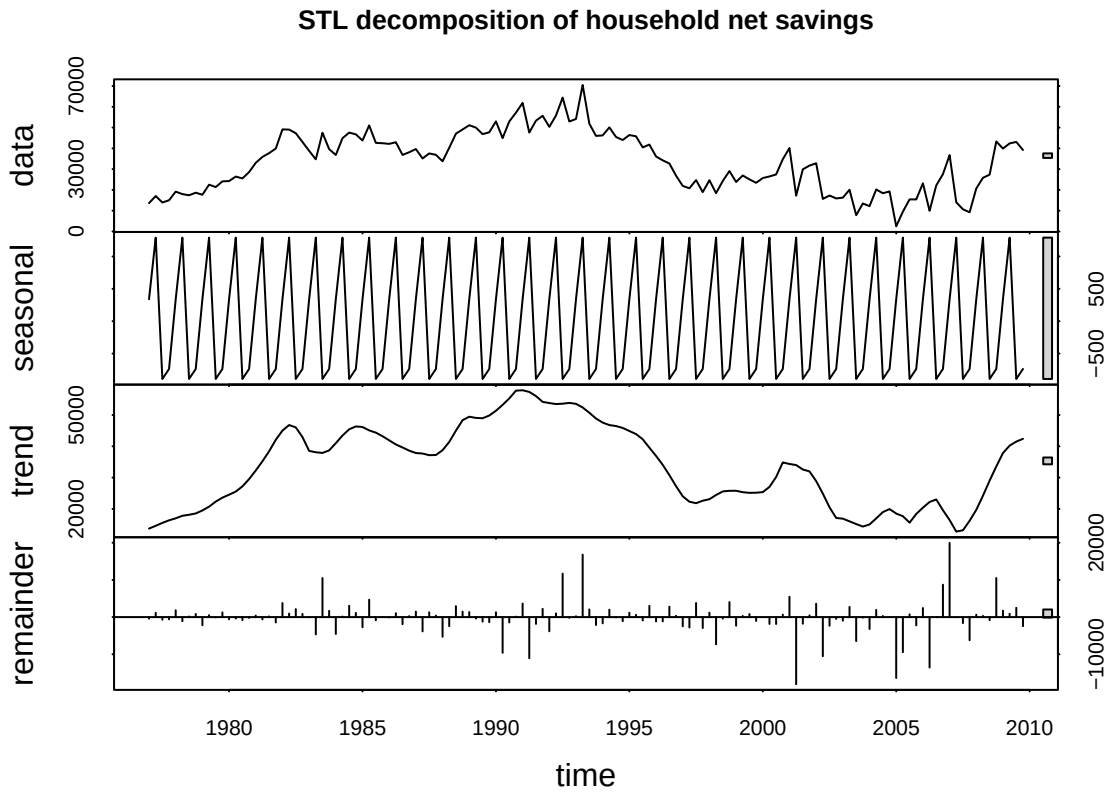
Table 1: Quarterly summary of household net savings

quarter_lab	mean_hh_save	sd_hh_save
Q1	33681.6	14169.5
Q2	32854.2	15258.4
Q3	33454.9	14494.1
Q4	33808.5	13364.3

Addition_2:

```
hh_ts <- ts(eda_df$hh_save, start = c(1977, 1), frequency = 4)

hh_stl <- stl(hh_ts, s.window = "periodic", robust = TRUE)
```

```r
plot(hh_stl, main = "STL decomposition of household net savings")
```




```r
stl_comp <- hh_stl$time.series

seasonal_strength <- max(
  0,
  1 - var(stl_comp[, "remainder"], na.rm = TRUE) /
    var(stl_comp[, "seasonal"] + stl_comp[, "remainder"], na.rm = TRUE)
)

trend_strength <- max(
  0,
  1 - var(stl_comp[, "remainder"], na.rm = TRUE) /
    var(stl_comp[, "trend"] + stl_comp[, "remainder"], na.rm = TRUE)
)

stl_strength_tbl <- tibble(
  component = c("seasonal_strength", "trend_strength"),
  value = c(seasonal_strength, trend_strength)
)

kable(stl_strength_tbl, digits = 3, caption = "STL component strength for household net savings")
```

Table 2: STL component strength for household net savings

component	value
seasonal_strength	0.000
trend_strength	0.891

The quarterly boxplot suggests only weak seasonal differences across Q1–Q4. This is consistent with the STL result, although STL extracts a periodic seasonal component, its estimated seasonal strength is essentially zero, indicating that seasonal variation is very small relative to the irregular component.

```r
hh_ts <- ts(eda_df$hh_save, start = c(1977, 1), frequency = 4)

auto_arima_hh <- auto.arima(hh_ts)

summary(auto_arima_hh)
```

```
## Series: hh_ts
## ARIMA(0,1,1)(0,0,2)[4]
##
## Coefficients:
##           ma1    sma1     sma2
##       -0.3112  0.0780  -0.1800
## s.e.   0.0956  0.0911   0.0972
##
## sigma^2 = 3.9e+07:  log likelihood = -1329.45
## AIC=2666.9   AICc=2667.22   BIC=2678.4
##
## Training set error measures:
##                    ME     RMSE      MAE       MPE     MAPE     MASE       ACF1
## Training set 258.7272 6149.934 4602.639 -6.933657 21.85228 0.616356 0.02424269
```

```r
cons_ts <- ts(eda_df$cons, start = c(1977, 1), frequency = 4)
gdi_ts <- ts(eda_df$gdi, start = c(1977, 1), frequency = 4)
gpdi_ts <- ts(eda_df$gpdi, start = c(1977, 1), frequency = 4)

auto_arima_cons <- auto.arima(cons_ts)
auto_arima_gdi <- auto.arima(gdi_ts)
auto_arima_gpdi <- auto.arima(gpdi_ts)

summary(auto_arima_cons)
```

```
## Series: cons_ts
## ARIMA(0,2,1)(0,0,1)[4]
##
## Coefficients:
##           ma1     sma1
##       -0.7207  -0.3669
## s.e.   0.0664   0.0878
##
## sigma^2 = 17855338:  log likelihood = -1269.56
## AIC=2545.12   AICc=2545.31   BIC=2553.73
##
## Training set error measures:
##                    ME     RMSE      MAE        MPE      MAPE      MASE
```

```
## Training set 137.1593 4161.042 3233.812 0.02890196 0.5178794 0.1696069
##                      ACF1
## Training set -0.04018861
```

```r
summary(auto_arima_gdi)
```

```
## Series: gdi_ts
## ARIMA(2,1,0)(1,0,0)[4] with drift
##
## Coefficients:
##          ar1      ar2     sar1      drift
##       0.7352  -0.3579  -0.2052  10739.040
## s.e.  0.0815   0.0822   0.0866   1223.611
##
## sigma^2 = 112804662:  log likelihood = -1398.74
## AIC=2807.48   AICc=2807.96   BIC=2821.85
##
## Training set error measures:
##                     ME     RMSE      MAE       MPE      MAPE      MASE
## Training set -14.51125 10417.86 6242.402 -0.145061 0.7323168 0.1318399
##                    ACF1
## Training set 0.02995205
```

```r
summary(auto_arima_gpdi)
```

```
## Series: gpdi_ts
## ARIMA(0,1,1)
##
## Coefficients:
##          ma1
##       0.3357
## s.e.  0.0745
##
## sigma^2 = 96202215:  log likelihood = -1389.46
## AIC=2782.91   AICc=2783.01   BIC=2788.66
##
## Training set error measures:
##                    ME     RMSE      MAE      MPE     MAPE      MASE       ACF1
## Training set 1112.822 9733.684 7494.871 0.441234 3.506758 0.3737059 0.01454896
```

However, all three series are non-stationary in levels, which motivates the need for differencing or other transformations to achieve stationarity before modeling.

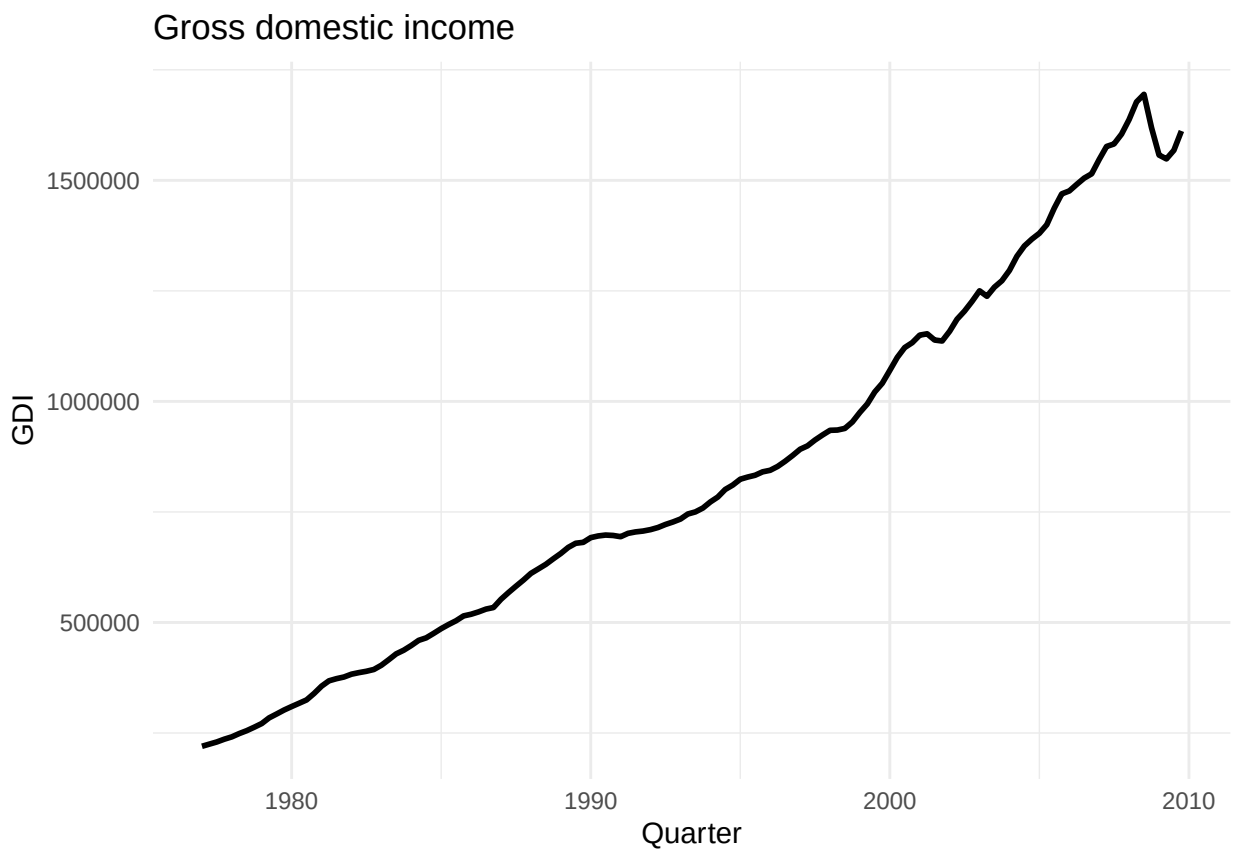
Transformations and Stationarity Checks

```r
eda_df <- eda_df |>
  mutate(
    gdi_qoq = (gdi / lag(gdi) - 1),
    gdi_yoy = (gdi / lag(gdi, 4) - 1),
    ldgdi = log(gdi) - log(lag(gdi))
  )

summary(select(eda_df, gdi, gdi_qoq, gdi_yoy, ldgdi))
```

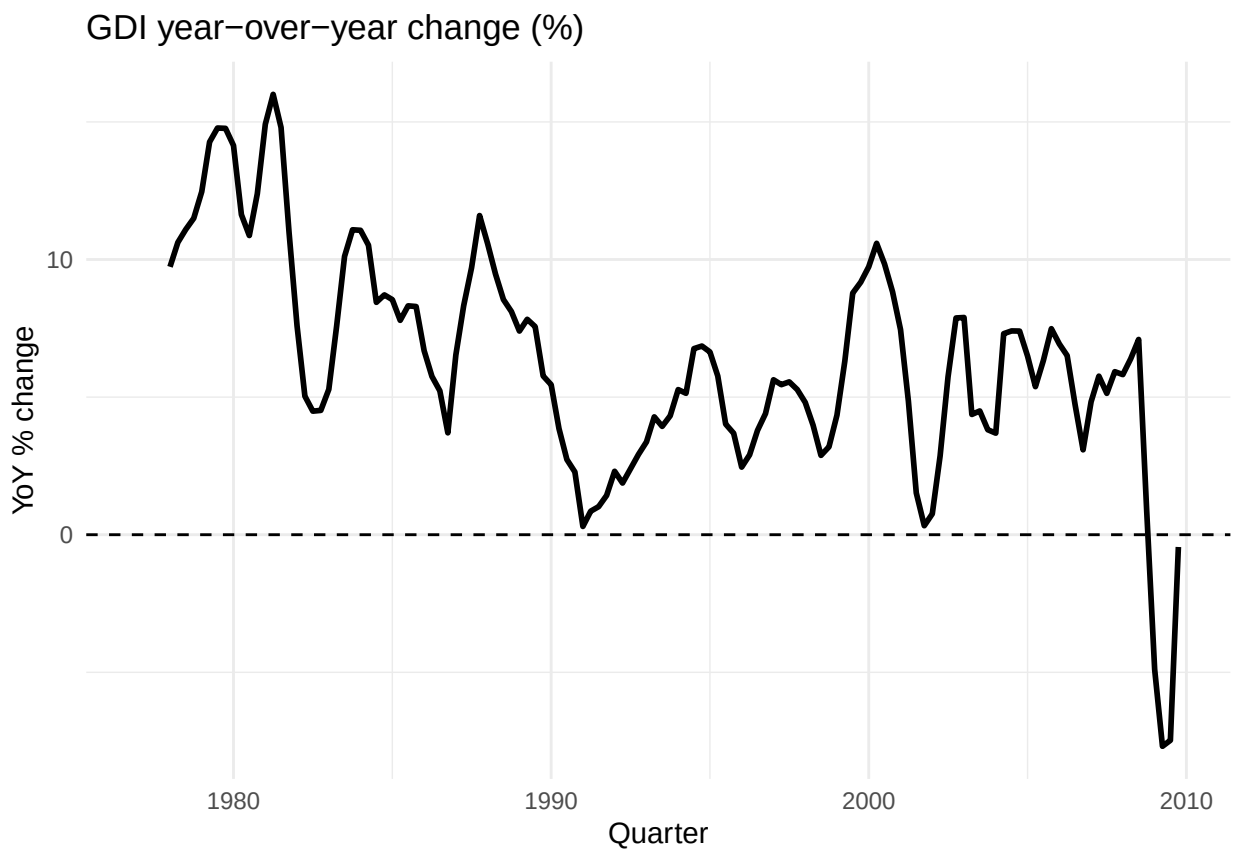
```
##       gdi             gdi_qoq             gdi_yoy             ldgdi
##  Min.   : 219912   Min.   :-0.044617   Min.   :-0.07688   Min.   :-0.04564
```

```
##  1st Qu.: 493038   1st Qu.: 0.009394   1st Qu.: 0.03969   1st Qu.: 0.00935
##  Median : 747714   Median : 0.015701   Median : 0.05871   Median : 0.01558
##  Mean   : 836060   Mean   : 0.015399   Mean   : 0.06302   Mean   : 0.01520
##  3rd Qu.:1150658   3rd Qu.: 0.022760   3rd Qu.: 0.08543   3rd Qu.: 0.02251
##  Max.   :1694512   Max.   : 0.049035   Max.   : 0.16002   Max.   : 0.04787
##                    NA's   :1           NA's   :4          NA's   :1
```

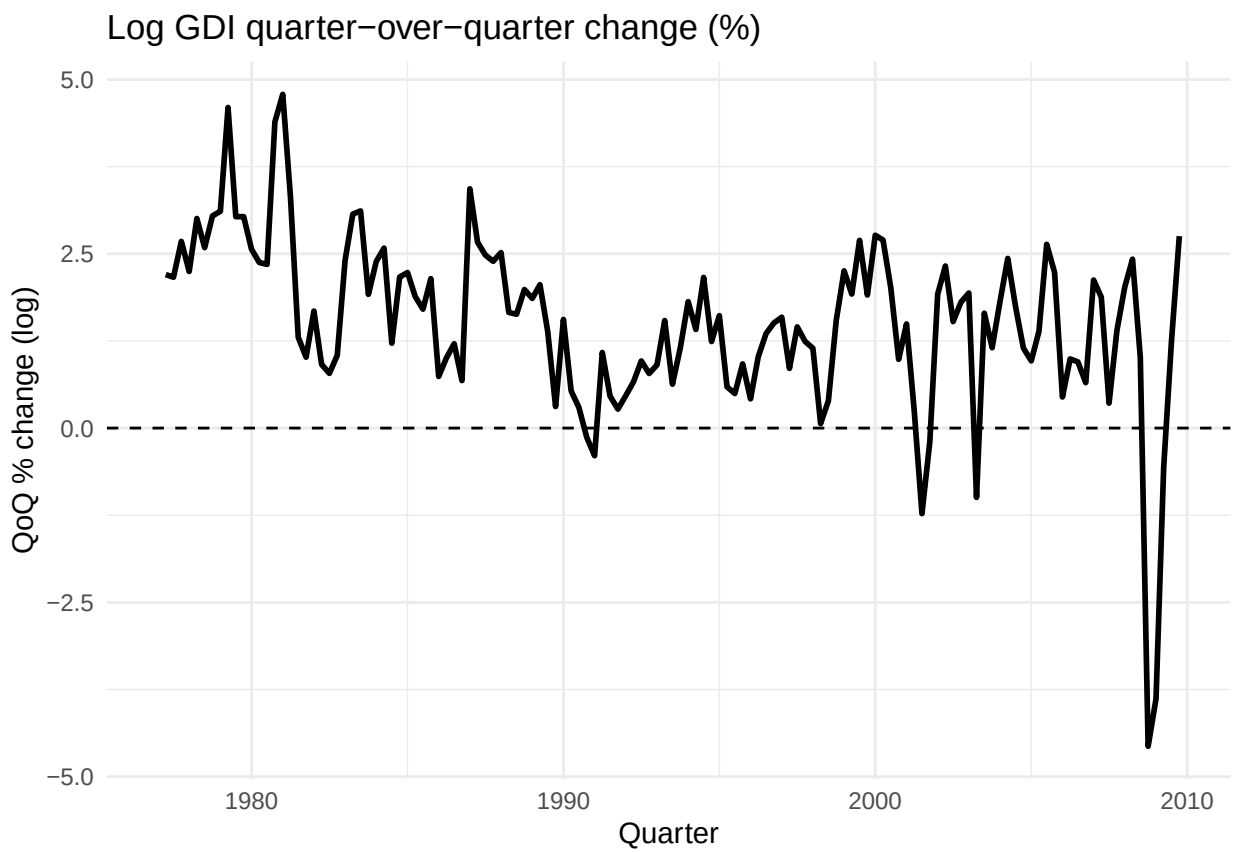
```r
p1 <- ggplot(eda_df, aes(x = date, y = gdi)) +
  geom_line(linewidth = 1) +
  theme_minimal() +
  labs(
    title = "Gross domestic income",
    x = "Quarter",
    y = "GDI"
  )

p2 <- ggplot(eda_df, aes(x = date, y = gdi_yoy*100)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_minimal() +
  labs(
    title = "GDI year-over-year change (%)",
    x = "Quarter",
    y = "YoY % change"
  )

p3 <- ggplot(eda_df, aes(x = date, y = ldgdi*100)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  theme_minimal() +
  labs(
    title = "Log GDI quarter-over-quarter change (%)",
    x = "Quarter",
    y = "QoQ % change (log)"
  )

p1
```


Gross domestic income
1500000
1000000
500000
GDI
1980
1990
2000
2010
Quarter

```
p2
```


GDI year−over−year change (%)
10
0
YoY % change
1980
1990
2000
2010
Quarter

```
p3
```

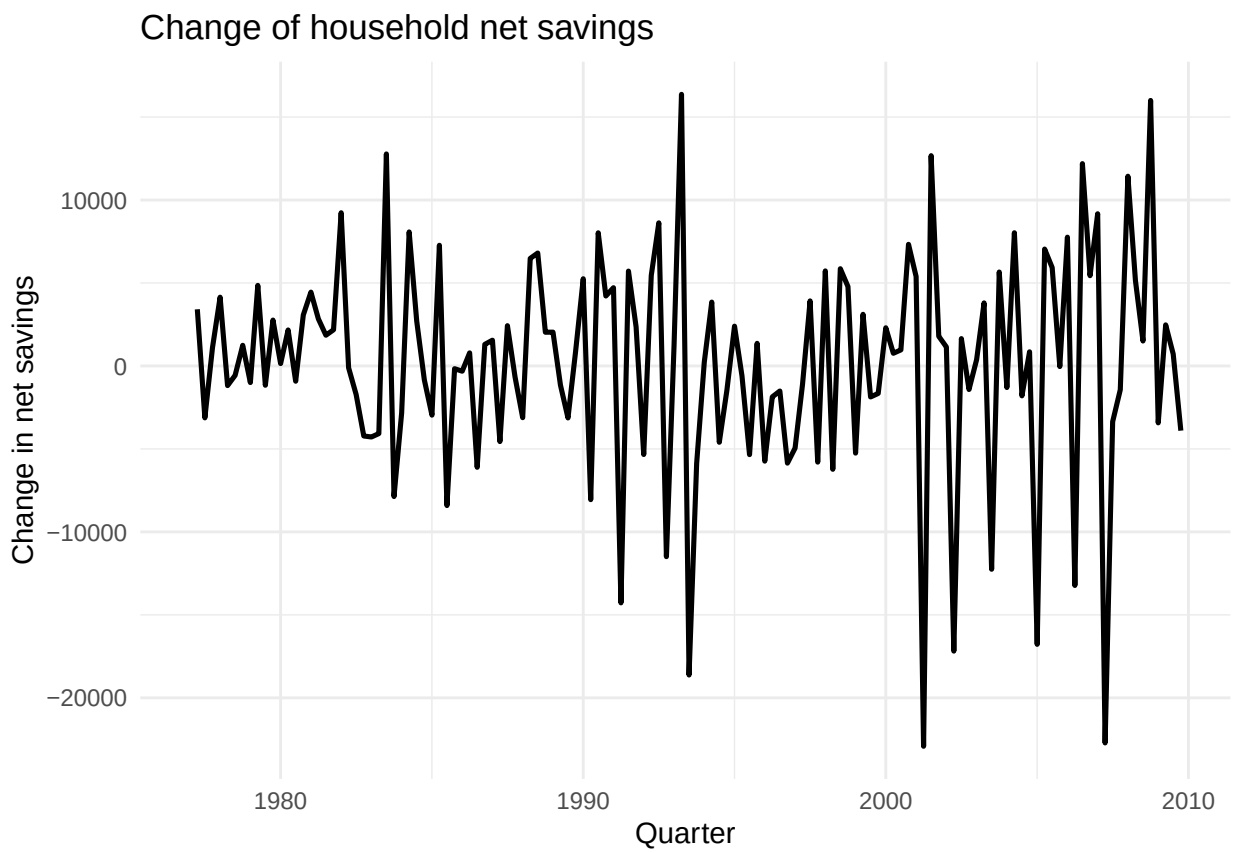



```r
eda_df <- eda_df |>
  mutate(
    dhh = hh_save - lag(hh_save),
    dcorp = corp_save - lag(corp_save),
    dnat = nat_save - lag(nat_save),
    dhh_rate = hh_rate - lag(hh_rate),
    dnat_rate = nat_rate - lag(nat_rate),
    dunemp = unemp - lag(unemp),
    dinterest = interest - lag(interest),
    # Use standard log differences for growth-style transformations.
    ldcons = log(cons) - log(lag(cons)),
    ldcpi = log(cpi) - log(lag(cpi)),
    ldgpdi = log(gpdi) - log(lag(gpdi))
  )

summary(select(
  eda_df,
  dhh,
  dcorp,
  dhh_rate,
  dnat_rate,
  dunemp,
  dinterest,
  ldcons,
  ldcpi,
  ldgpdi
```

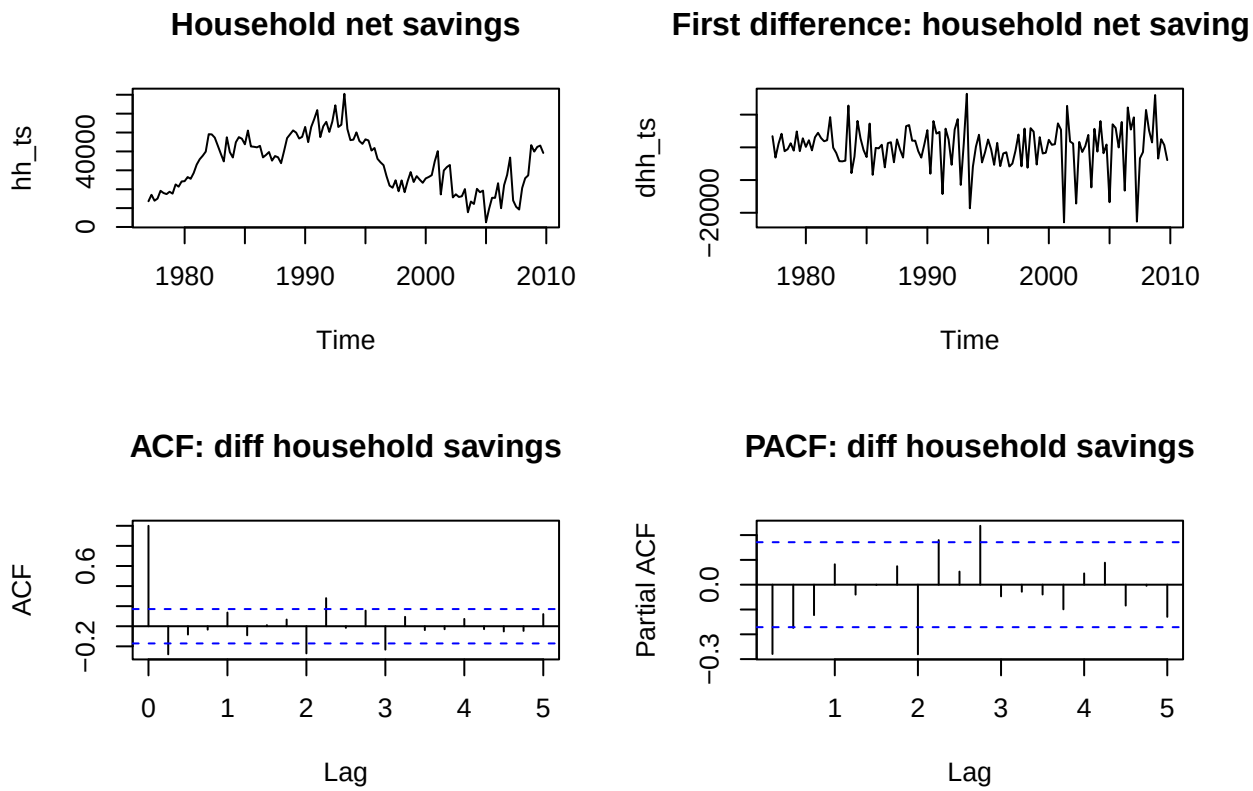
```
))
```

```
##       dhh               dcorp             dhh_rate
##  Min.   :-22920.0   Min.   :-53876.0   Min.   :-0.0254234
##  1st Qu.: -3040.0   1st Qu.: -2762.0   1st Qu.:-0.0059458
##  Median :   852.0   Median :  1072.0   Median : 0.0003992
##  Mean   :   195.1   Mean   :   133.8   Mean   :-0.0002875
##  3rd Qu.:  4180.0   3rd Qu.:  4202.0   3rd Qu.: 0.0047289
##  Max.   : 16368.0   Max.   : 24532.0   Max.   : 0.0272227
##  NA's   :1          NA's   :1          NA's   :1
##    dnat_rate              dunemp            dinterest           ldcons
##  Min.   :-0.0428550   Min.   :-1.000000   Min.   :-3.6200   Min.   :-0.021070
##  1st Qu.:-0.0050400   1st Qu.:-0.200000   1st Qu.:-0.5257   1st Qu.: 0.002112
##  Median : 0.0003252   Median :-0.100000   Median : 0.0000   Median : 0.007701
##  Mean   :-0.0003667   Mean   : 0.006107   Mean   :-0.0585   Mean   : 0.006610
##  3rd Qu.: 0.0050821   3rd Qu.: 0.200000   3rd Qu.: 0.4429   3rd Qu.: 0.011744
##  Max.   : 0.0220223   Max.   : 1.800000   Max.   : 3.4486   Max.   : 0.019634
##  NA's   :1            NA's   :1           NA's   :1         NA's   :1
##      ldcpi               ldgpdi
##  Min.   :-0.020962   Min.   :-0.157243
##  1st Qu.: 0.003965   1st Qu.:-0.018941
##  Median : 0.008253   Median : 0.011659
##  Mean   : 0.009563   Mean   : 0.006758
##  3rd Qu.: 0.013899   3rd Qu.: 0.036709
##  Max.   : 0.038221   Max.   : 0.109470
##  NA's   :1           NA's   :1
```

```r
diff_long <- eda_df |>
  select(date, dhh) |>
  pivot_longer(
    cols = -date,
    names_to = "series",
    values_to = "value"
  )
ggplot(diff_long, aes(x = date, y = value)) +
  geom_line(linewidth = 0.9) +
  theme_minimal() +
  labs(
    title = "Change of household net savings",
    x = "Quarter",
    y = "Change in net savings"
  )
```




```
hh_ts <- ts(
  eda_df$hh_save,
  start = c(1977, 1),
  frequency = 4
)

dhh_ts <- diff(hh_ts)

par(mfrow = c(2, 2))
plot(hh_ts, main = "Household net savings")
plot(dhh_ts, main = "First difference: household net savings")
acf(na.omit(dhh_ts), lag.max = 20, main = "ACF: diff household savings")
pacf(na.omit(dhh_ts), lag.max = 20, main = "PACF: diff household savings")
```

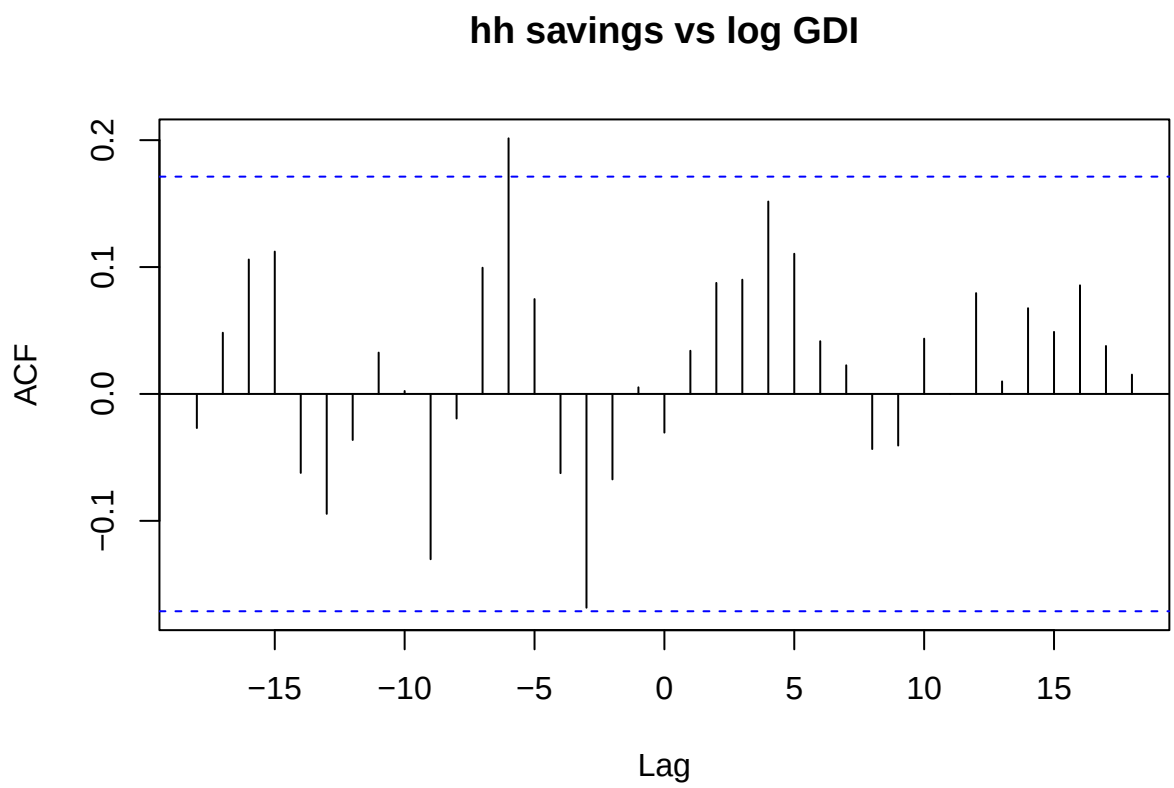



```r
par(mfrow = c(1, 1))

adf.test(na.omit(dhh_ts))
```

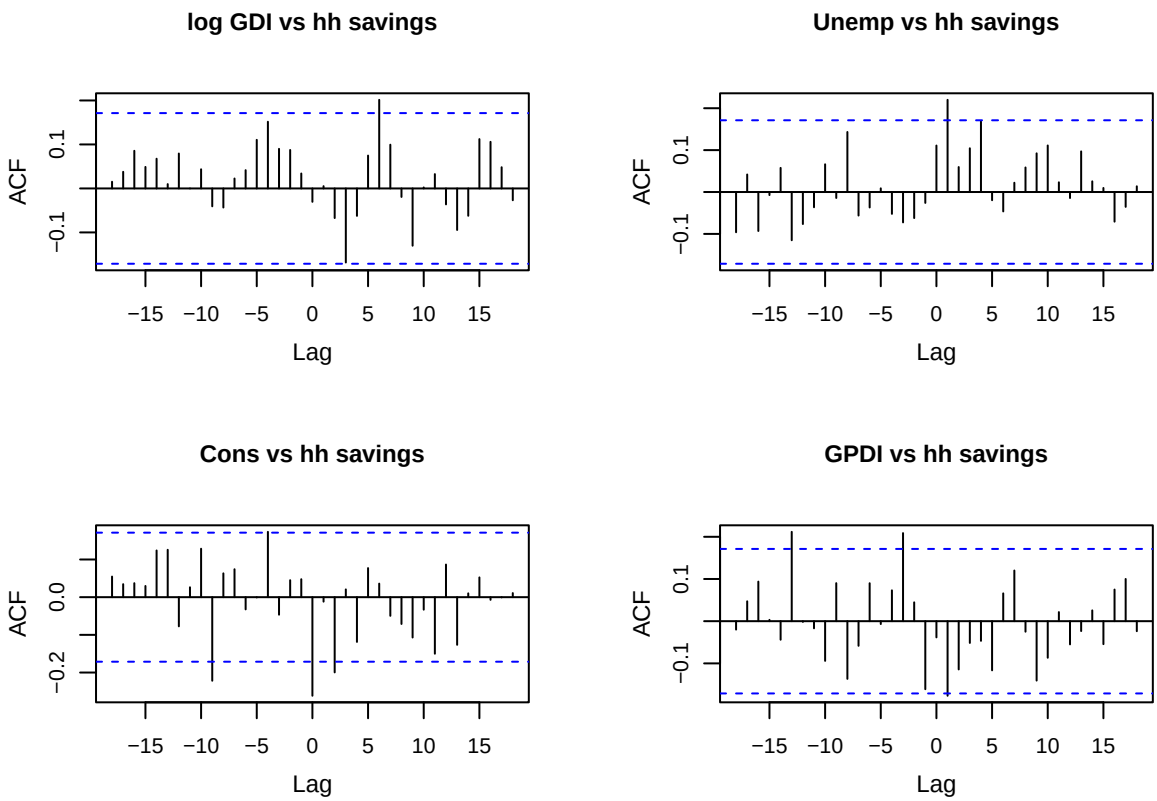
```
##
##  Augmented Dickey-Fuller Test
##
## data:  na.omit(dhh_ts)
## Dickey-Fuller = -4.9114, Lag order = 5, p-value = 0.01
## alternative hypothesis: stationary
```

- From the above, we see that the differenced household net savings appear stationary based on the ADF test and the visual diagnostics.
- But they don't demonstrate strong autocorrelation, which suggests that they may be mostly driven by exogenous factors rather than strong internal dynamics.

```r
ccf(na.omit(eda_df$dhh), na.omit(eda_df$ldgdi),
    main = "hh savings vs log GDI")
```




```r
par(
  mfrow = c(2, 2),
  mar = c(5, 5, 3, 1),
  mgp = c(2, 0.8, 0),
  cex.main = 0.95,
  cex.lab = 0.9,
  cex.axis = 0.85
)

ccf(na.omit(eda_df$ldgdi), na.omit(eda_df$dhh),
    main = "log GDI vs hh savings")

ccf(na.omit(eda_df$dunemp), na.omit(eda_df$dhh),
    main = "Unemp vs hh savings")

ccf(na.omit(eda_df$ldcons), na.omit(eda_df$dhh),
    main = "Cons vs hh savings")

ccf(na.omit(eda_df$ldgpdi), na.omit(eda_df$dhh),
    main = "GPDI vs hh savings")
```




```
par(mfrow = c(1, 1))
```

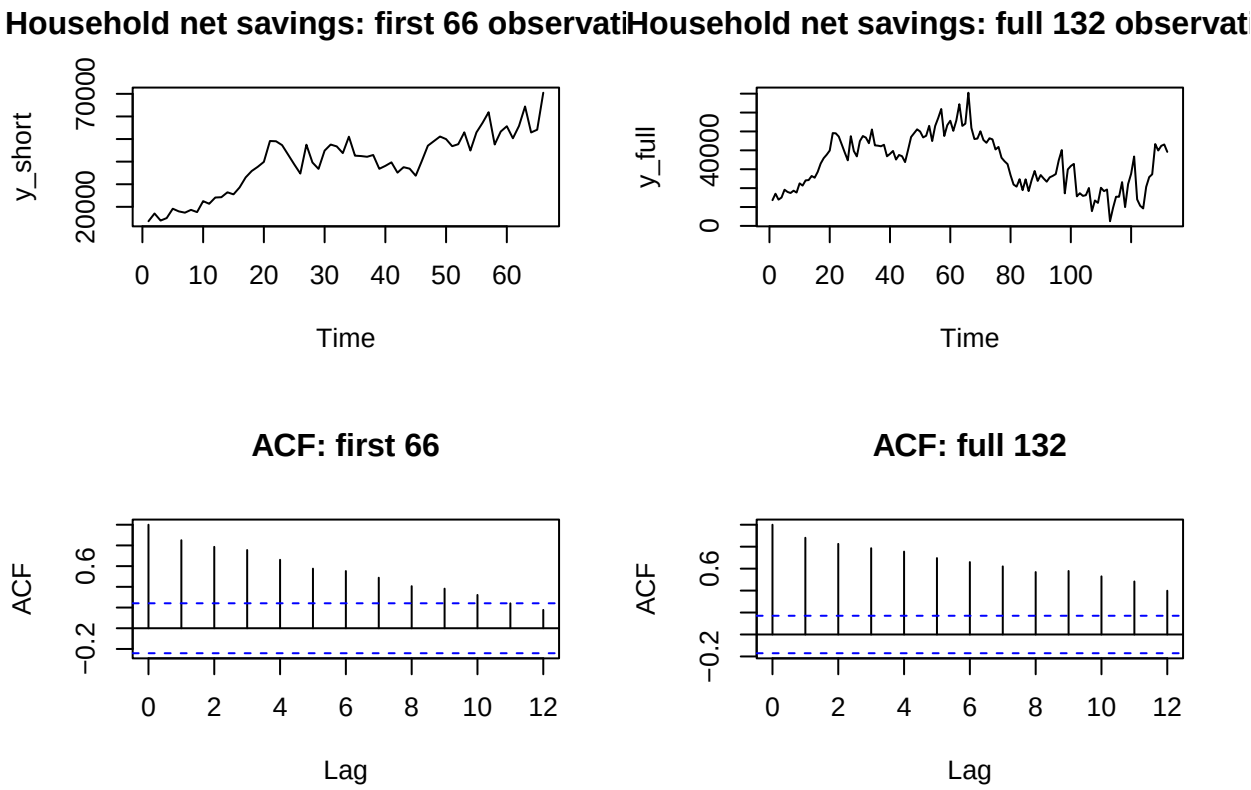
From the above, we see the following:

- Household net savings show strong contemporaneous correlation with GDI, consumption, and CPI changes, as well as some leading correlation with unemployment changes.
- GPDI shows a clear leading relationship with household savings at lags 1–3, which motivates its use as an exogenous regressor in ARIMAX models.

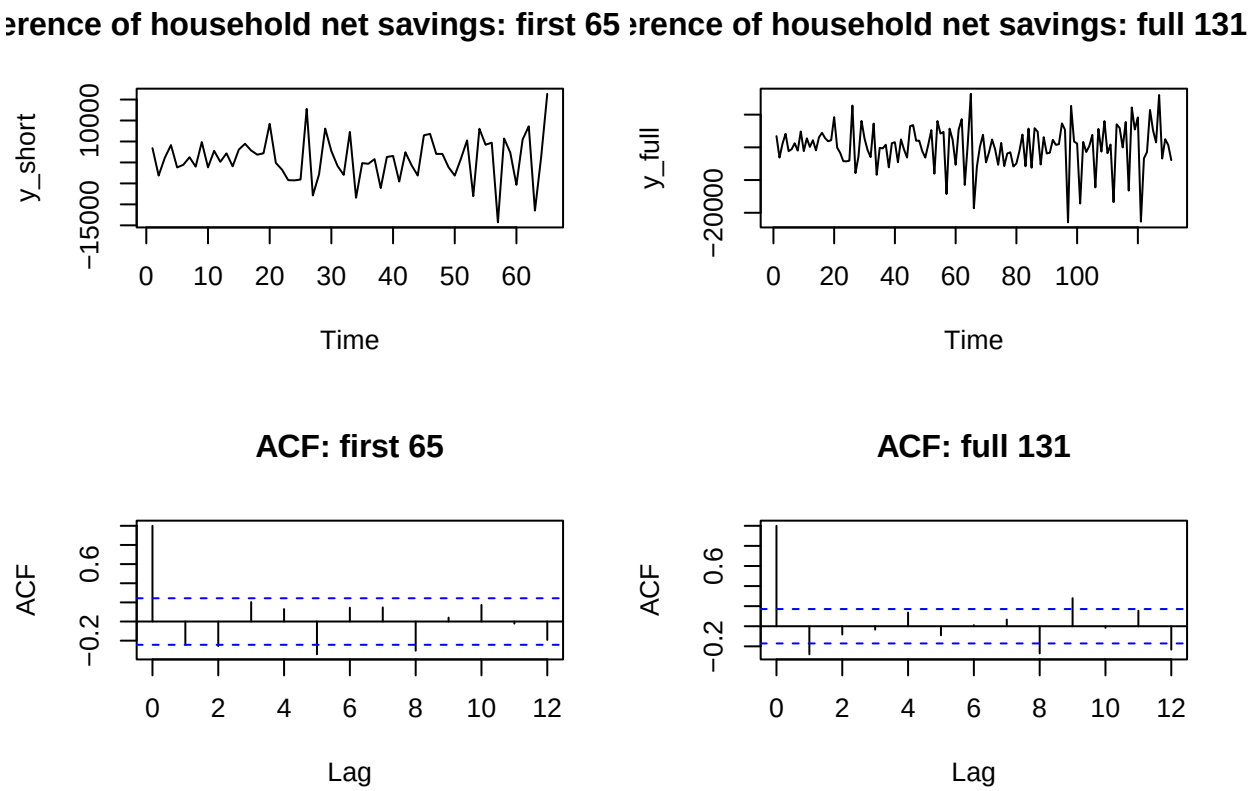
#Addition_3

```
compare_sample_acf_var <- function(y, lag.max = 12, split_frac = 0.5, series_name = "series") {
  y <- as.numeric(na.omit(y))

  n_full <- length(y)
  n_short <- floor(split_frac * n_full)

  y_short <- y[1:n_short]
  y_full  <- y

  par(mfrow = c(2, 2))
  plot.ts(y_short, main = paste0(series_name, ": first ", n_short, " observations"))
  plot.ts(y_full,  main = paste0(series_name, ": full ", n_full, " observations"))
  acf(y_short, lag.max = lag.max, main = paste0("ACF: first ", n_short))
  acf(y_full,  lag.max = lag.max, main = paste0("ACF: full ", n_full))
  par(mfrow = c(1, 1))

  out_tbl <- tibble(
    sample = c(paste0("first_", n_short), paste0("full_", n_full)),
    mean = c(mean(y_short), mean(y_full)),
```

```r
    variance = c(var(y_short), var(y_full)),
    sd = c(sd(y_short), sd(y_full))
  )

  kable(out_tbl, digits = 3,
        caption = paste0(series_name, ": mean and variance comparison across sample lengths"))

  invisible(out_tbl)
}

# Level series
level_compare_tbl <- compare_sample_acf_var(
  hh_ts,
  lag.max = 12,
  split_frac = 0.5,
  series_name = "Household net savings"
)
```

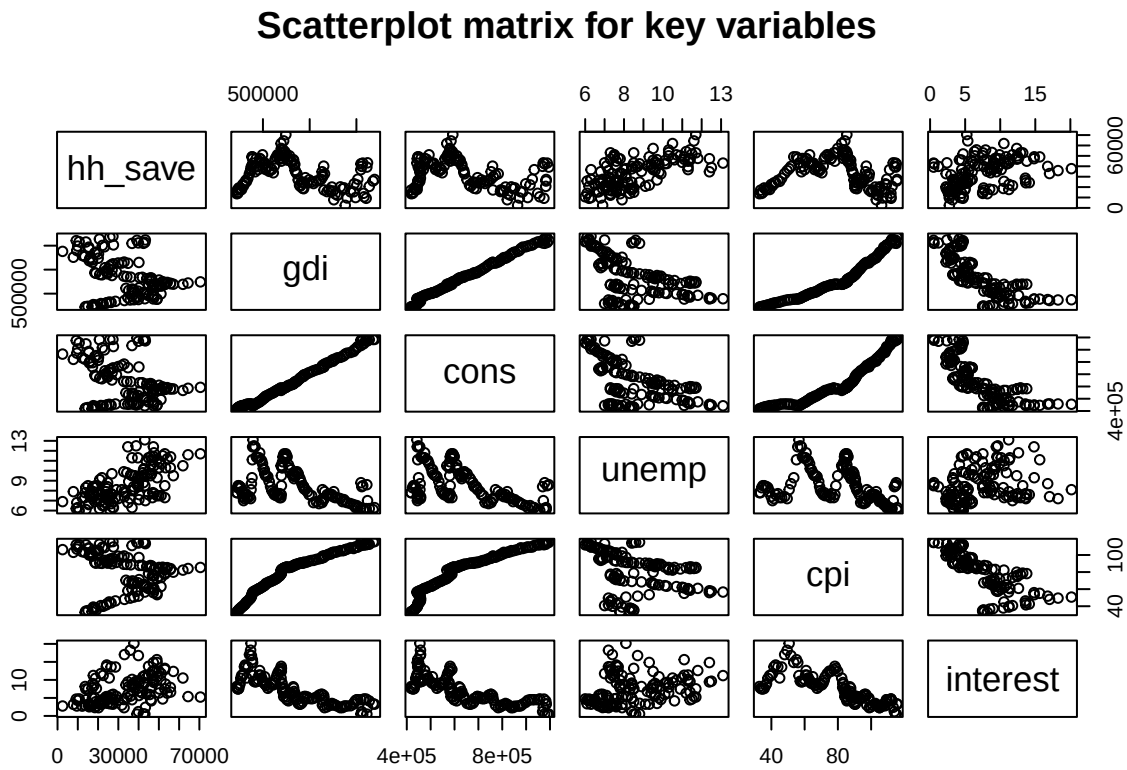



```r
# Differenced series
diff_compare_tbl <- compare_sample_acf_var(
  dhh_ts,
  lag.max = 12,
  split_frac = 0.5,
  series_name = "First difference of household net savings"
)
```

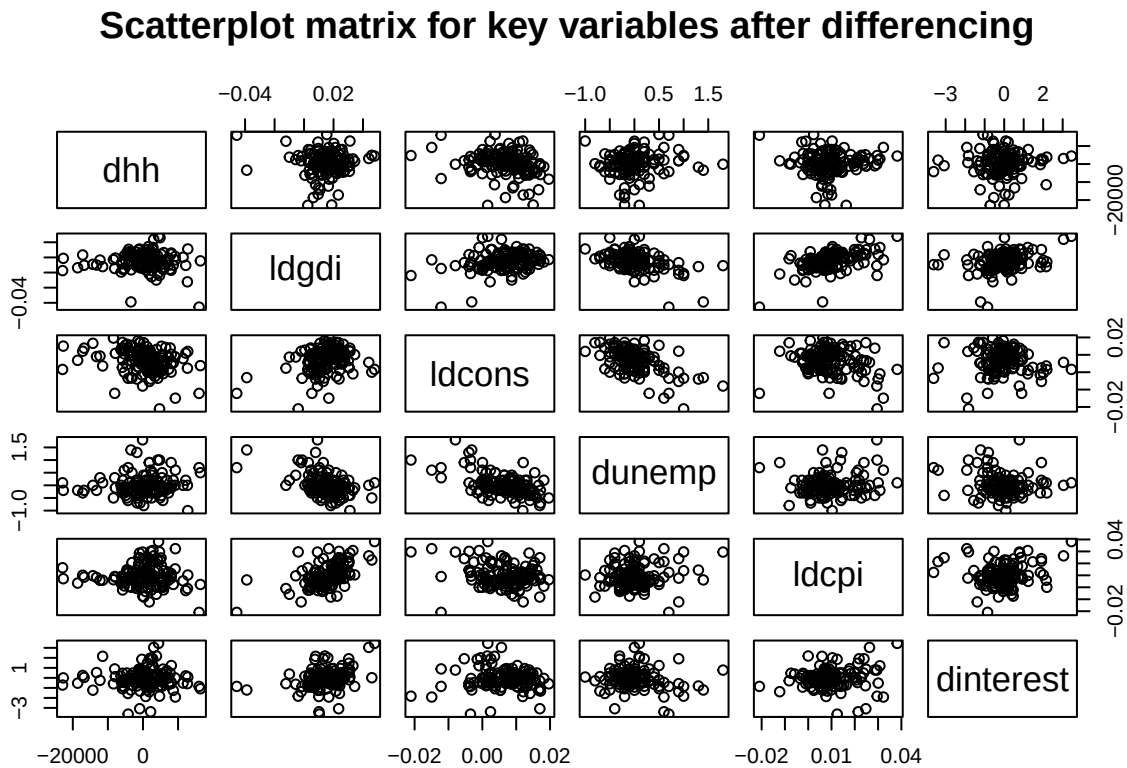


Comparing the first half of the sample with the full sample shows that household net savings in levels is not stable across sample lengths. Its ACF remains large and slowly decays, which is consistent with non-stationarity. After first differencing, the series fluctuates around a more stable level and the ACF becomes much smaller, supporting the use of differencing before ARIMA-type modeling.

```
pairs(
  eda_df |>
    select(hh_save, gdi, cons, unemp, cpi, interest) |>
    na.omit(),
  main = "Scatterplot matrix for key variables"
)
```




```r
pairs(
  eda_df |>
    select(dhh, ldgdi, ldcons, dunemp, ldcpi, dinterest) |>
    na.omit(),
  main = "Scatterplot matrix for key variables after differencing"
)
```



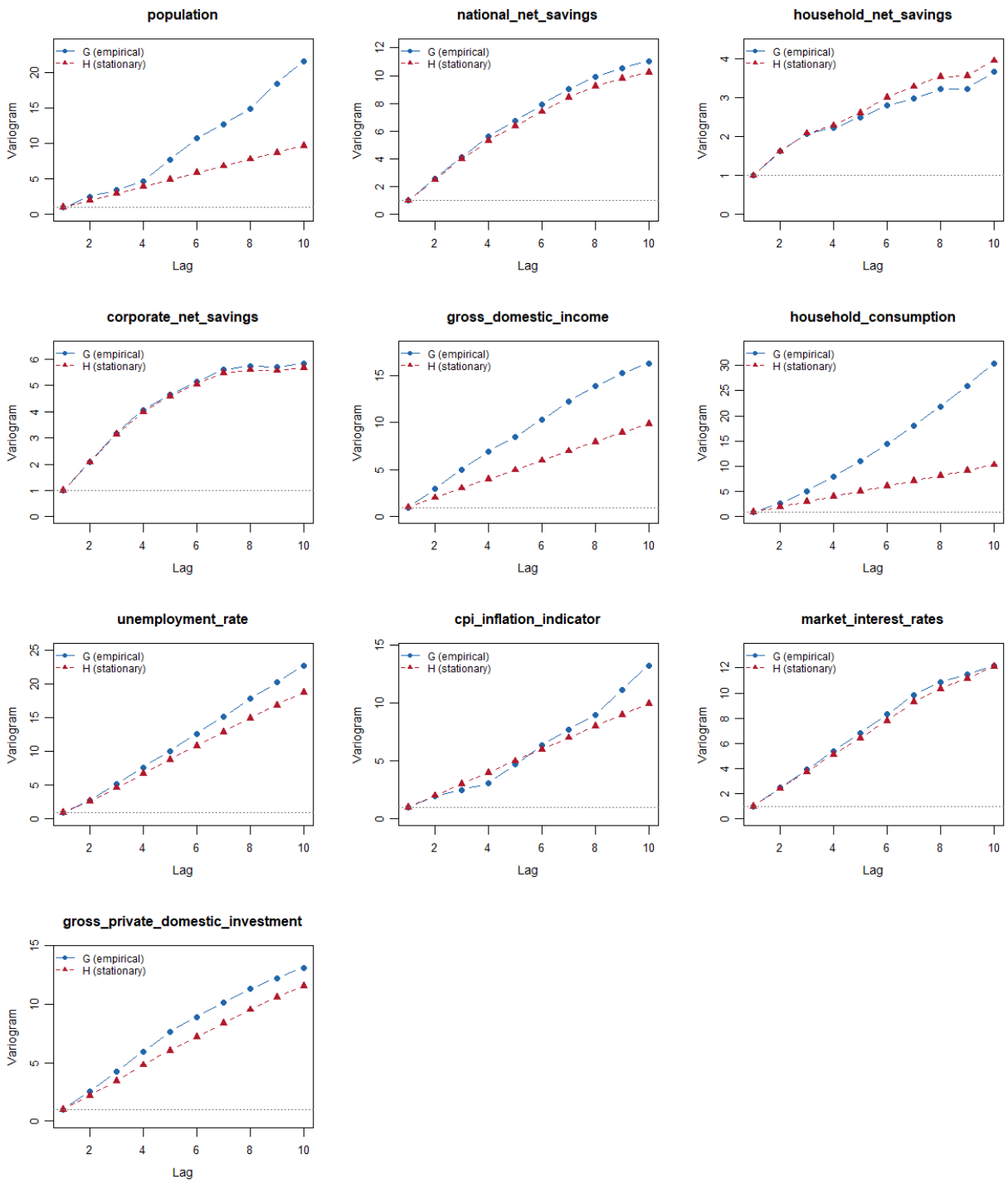

```r
glimpse(eda_df)
```

```
## Rows: 132
## Columns: 27
## $ date        <date> 1977-01-01, 1977-04-01, 1977-07-01, 1977-10-01, 1978-01-0~
## $ pop         <dbl> 23591713, 23656739, 23725843, 23801445, 23849503, 23903687~
## $ nat_save    <dbl> 15180, 14684, 13724, 14136, 13512, 14992, 15232, 15124, 16~
## $ hh_save     <dbl> 13632, 17048, 13920, 15024, 19168, 17984, 17416, 18660, 17~
## $ corp_save   <dbl> 4660, 3260, 3592, 4984, 3892, 3504, 4576, 3336, 4588, 6492~
## $ gdi         <dbl> 219912, 224808, 229724, 235960, 241320, 248684, 255204, 26~
## $ cons        <dbl> 418612, 417567, 424081, 428064, 431165, 434052, 436086, 43~
## $ unemp       <dbl> 7.8, 7.8, 8.3, 8.5, 8.5, 8.4, 8.4, 8.3, 7.9, 7.4, 7.0, 7.2~
## $ cpi         <dbl> 32.8, 33.5, 34.2, 34.9, 35.7, 36.6, 37.1, 37.9, 39.0, 39.9~
## $ interest    <dbl> 8.164062, 7.700000, 7.500000, 7.500000, 7.630769, 8.492308~
## $ gpdi        <dbl> 135625, 135843, 133541, 132462, 128906, 132350, 135470, 14~
## $ hh_rate     <dbl> 0.06198843, 0.07583360, 0.06059445, 0.06367181, 0.07942980~
## $ nat_rate    <dbl> 0.06902761, 0.06531796, 0.05974125, 0.05990846, 0.05599204~
## $ quarter_lab <fct> Q1, Q2, Q3, Q4, Q1, Q2, Q3, Q4, Q1, Q2, Q3, Q4, Q1, Q2, Q3~
## $ gdi_qoq     <dbl> NA, 0.022263451, 0.021867549, 0.027145618, 0.022715715, 0.~
## $ gdi_yoy     <dbl> NA, NA, NA, NA, 0.09734803, 0.10620618, 0.11091571, 0.1149~
## $ ldgdi       <dbl> NA, 0.022019238, 0.021631884, 0.026783711, 0.022461554, 0.~
## $ dhh         <dbl> NA, 3416, -3128, 1104, 4144, -1184, -568, 1244, -1000, 485~
## $ dcorp       <dbl> NA, -1400, 332, 1392, -1092, -388, 1072, -1240, 1252, 1904~
## $ dnat        <dbl> NA, -496, -960, 412, -624, 1480, 240, -108, 908, 5572, 221~
## $ dhh_rate    <dbl> NA, 0.013845168, -0.015239148, 0.003077356, 0.015757994, -~
## $ dnat_rate   <dbl> NA, -0.0037096508, -0.0055767055, 0.0001672043, -0.0039164~
```

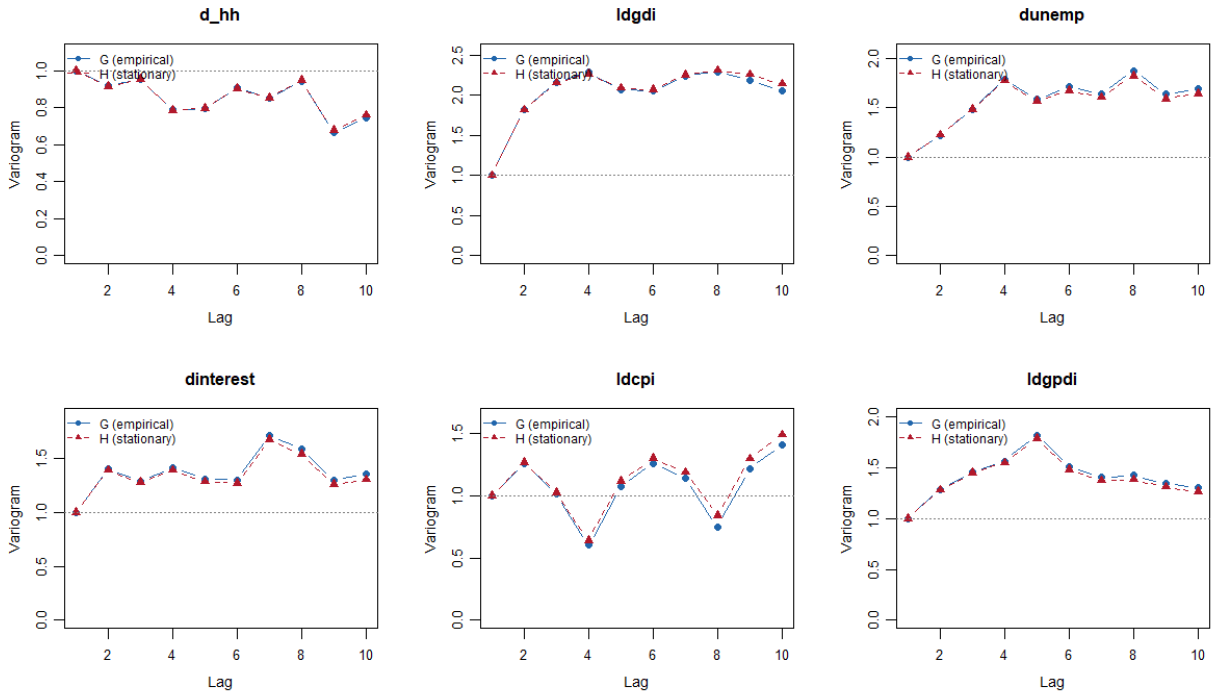
```
## $ dunemp      <dbl> NA, 0.0, 0.5, 0.2, 0.0, -0.1, 0.0, -0.1, -0.4, -0.5, -0.4,~
## $ dinterest   <dbl> NA, -0.46406250, -0.20000000, 0.00000000, 0.13076923, 0.86~
## $ ldcons      <dbl> NA, -2.499466e-03, 1.547946e-02, 9.348242e-03, 7.218130e-0~
## $ ldcpi       <dbl> NA, 0.02111692, 0.02068021, 0.02026119, 0.02266386, 0.0248~
## $ ldgpdi      <dbl> NA, 0.001606083, -0.017091260, -0.008112735, -0.027212355,~
```

```r
saveRDS(eda_df, file = "../output/macro_df_diff.rds")
```

```r
knitr::include_graphics("../output/variograms_levels.png")
```




```
knitr::include_graphics("../output/variograms_differenced.png")
```



- The plots indicate that most variables in levels are not stationary, because their empirical variogram G increases with lag and often departs from the theoretical stationary variogram H. - For many level variables, this increase is approximately linear rather than sharply curved, which suggests that a single difference is likely sufficient to achieve approximate stationarity. This pattern is especially obvious for variables such as national net savings, household net savings, corporate net savings, unemployment rate, market interest rates, household net saving rates, and corporate net saving rates, where G rises with lag and H does not fully overlap with it. - Some series, such as population, gross domestic income, household consumption, CPI, and gross private domestic investment, show stronger divergence between G and H, with growth in G that is steeper. This indicates these variables are even less stationary. In contrast, the differenced variables (`dhh`, `dcorp`, `ldgdi`, `dunemp`, `dinterest`, `ldcpi`, `ldgpdi`) and their lagged versions show overlapping G and H. This strongly suggests that these transformed series are approximately stationary. Overall, the variograms support the conclusion that first differencing is generally needed for the level variables, while there is little evidence that second differencing is required.